

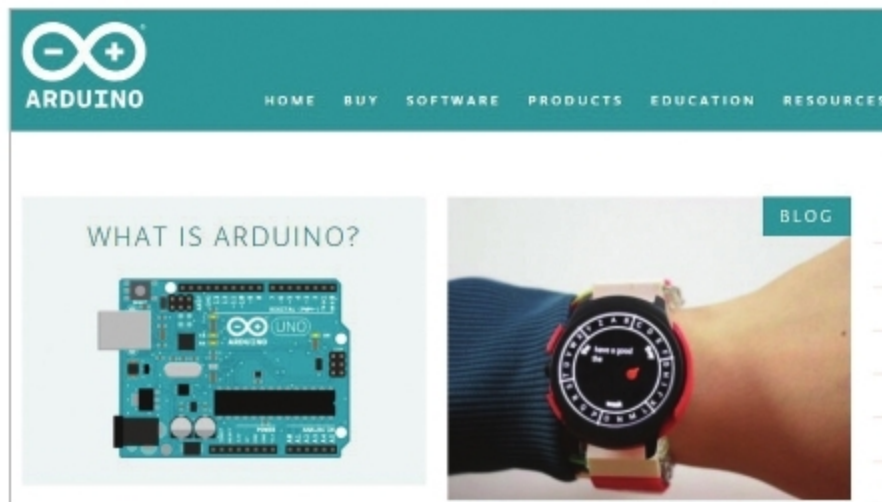


Figure 2: The Arduino board being used in wearable devices
(Source: Official portal of Arduino)

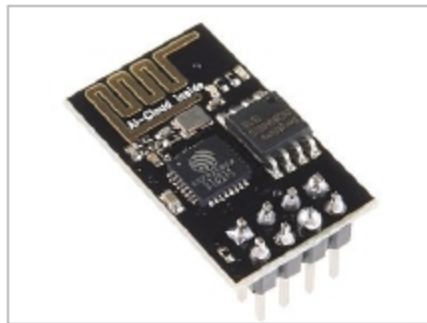


Figure 3: ESP8266 microchip with Wi-Fi connectivity



Figure 4: NodeMCU firmware for the Internet of Things (IoT)

the base programming languages used to program and customise the behaviour of the Arduino microcontroller board. The Arduino IDE can be downloaded without any cost, along with the live updates of the software.

Preparing Arduino for Internet of Things (IoT) applications

Arduino can be connected with live platforms of Internet of Things (IoT) by using Wi-Fi or similar network based chips. ESP8266 is a very popular device for communication with the Internet that can be easily interfaced with Arduino microcontroller boards. The ESP8266 Wi-Fi chip is very low cost and can be directly connected with the Arduino board to establish communication with the real cloud.

NodeMCU, an open source IoT based hardware platform with inbuilt Wi-Fi: To communicate and transfer data to the cloud in real-time, there is another chip called NodeMCU (<http://www.nodemcu.com>) that can directly communicate with live servers. NodeMCU integrates the

firmware with inbuilt Wi-Fi connectivity so that direct interaction with the network can be done without physical jumper wires. It is an Arduino-like board that is used for interfacing with the cloud in real-time so that the live signals or data can be logged or stored for predictive analysis.

Storing sensor data in IoT platforms

There are a number of IoT platforms that can be used for the storage of real-time streaming sensor data from Arduino and similar boards. Once the circuit is created using Arduino, the signals can be transmitted and stored directly onto IoT platforms. These open source IoT platforms can be used for storage, processing, prediction and visualisation of the data recorded from the sensors on the Arduino board.

A few prominent IoT platforms that can be used for device management and the visualisation of data are:

- ThingSpeak: thingspeak.com
- Carriots: carriots.com
- KAA: kaaproject.org
- Things Board: thingsboard.io
- MainFlux: mainflux.com
- Thinger: thinger.io
- DeviceHive: devicehive.com

Transferring data to ThingSpeak

ThingSpeak (<https://www.thingspeak.com>) is a widely used open IoT platform for the collection, storage and visualisation of real-time sensor data fetched from Arduino, Raspberry Pi, BeagleBone Black and similar boards. Live streaming data on the cloud can be preserved using channels in ThingSpeak.

To work with ThingSpeak, a free account can be created. After successfully signing up on ThingSpeak, a channel is created on the dashboard and a 'Write Key' is generated, which is to be used in the Arduino code.

The key features of ThingSpeak include the following:

- Configuration of devices for sending data to a cloud based environment
- Aggregation of sensor data from different devices and motherboards connected with sensors
- Visualisation of real-time sensor data including historical records
- Pre-processing and evaluation of sensor data for predictive analysis and knowledge discovery

The source code that can be customised and executed on Arduino IDE so that real-time data can be fetched on board and then transmitted to ThingSpeak is given below. For simplicity, the execution of the code is done for NodeMCU hardware, which has inbuilt Wi-Fi for sending the data to the IoT platform.

```
#include <ESP8266WiFi.h>
const char* mynetwork      = "<Wi-Fi Network Name>";
const char* mypassword     = "<Wi-Fi Network Mypassword>";
const char* thingspeakhost = "api.thingspeak.com";
```



Figure 5: ThingSpeak: An open source platform for the Internet of Things (IoT)